MICHAEL LO RUSSO

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SUMMARY

Engineering student at UNSW specialising in Mechatronics, with practical experience spanning embedded systems, mechanical design, software development, and construction. Built a residential house, autonomous robots, and 3D-printed thermal hardware. **Available for full-time work while completing my degree.**

Visit my website portfolio for more details on my projects: **mlorusso.com**

EDUCATION

Bachelor of Engineering (Honours) (Robotics and Mechatronics) Jan 2022 – Aug 2026
UNSW | WAM: Distinction

WORK EXPERIENCE

Research Intern, Honours Thesis | *Coolsheet partnering with UNSW* Sep 2025 – Present

- Developing a web-based modelling tool (Python back-end, HTML/CSS front-end) that simulates energy performance and cost-saving potential of Coolsheet's photovoltaic-thermal (PVT) panels and heat pump systems for commercial and industrial applications.
- Delivered a working v1 calculator with modular architecture, integrating Australian climate data via weather APIs to generate location-aware performance estimates across multiple building types.
- Cross-validating simulation outputs against real-world installation data from Coolsheet's commercial PVT deployments.

Experience Consultant | *Samsung Electronics (Parramatta)* Jan 2023 – Present

- Diagnosed and resolved hardware/software issues across Samsung mobile, tablet, and smart device lines, consistently maintaining a net promoter score above 85%.
- Generated $950k+ in sales revenue while communicating complex technical information to both technical and non-technical customers across in-store and phone channels.
- Trained new staff on product knowledge and store procedures. Managed daily store opening and closing operations and assigned sales budgets to team members.

ENGINEERING PROJECTS

Rubik's Cube Solving Robot Mar 2026 – May 2026

- Engineered a 5-axis robotic manipulator (NEMA 17 steppers, TMC2209, ESP32) that accepts a scrambled cube state via a custom web interface and computes a solution in under 1 second using the Kociemba two-phase algorithm.
- Integrated full-stack system spanning Fusion 360 mechanical design, C++ firmware with stepper motor control, a Python solving server, and an HTML/JavaScript UI for cube face input.

Micromouse Maze Navigation Robot | *UNSW* May 2024 – Aug 2024

- Designed and programmed an autonomous maze-solving robot using LiDAR, IMU, and wheel encoders with PID control; navigated a 3.6 × 2.4 m arena with walls and obstacles in under 90 seconds.
- Implemented BFS path planning with OpenCV occupancy grid mapping on an Arduino Nano, processing sensor data in real time for autonomous navigation.

Residential Construction Jun 2024 – Apr 2025

- Constructed a residential house alongside my father, executing framing, roofing, plumbing, and electrical work while ensuring all stages adhered to building compliance and regulatory standards.

Custom GPU Cooling Funnels Jan 2025

- Designed and 3D-printed airflow funnels in Fusion 360 across 3 design iterations; reduced GPU temperatures by 7°C under full load, inspired by automotive ducting principles.

UR5e Robotic System | *UNSW* May 2025 – Aug 2025

- Programmed a UR5e industrial robot for autonomous math problem solving, pick-and-place, and orientation adjustment. Developed trajectory planning to produce smooth pen lifts and legible handwriting output.

Unmanned Ground Vehicle Control System | *UNSW* Sep 2024 – Dec 2024

- Built a multi-threaded C++ control system integrating LiDAR, GNSS, Xbox controller, and collision sensors for autonomous and teleoperated driving with real-time crash avoidance and MATLAB visualisation.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, VHDL, HTML/CSS
Systems & Control: Sensor fusion (LiDAR, GNSS, IMU), PID control, real-time path planning, OpenCV, Arduino, MATLAB/Simulink
CAD & Manufacturing: SolidWorks, Fusion 360, Bambu Studio, 3D printing (FDM/PLA)
Web & Tools: Next.js, TypeScript, Tailwind CSS, Git, VS Code, LaTeX
AI & Automation: Claude Code, Codex, Cursor, Vercel, Antigravity, n8n